



Department of Computer Science

National University of Computer and Emerging Sciences
Lahore Campus

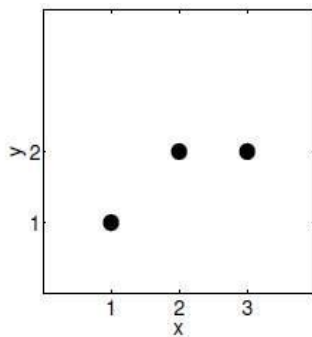
Data Mining

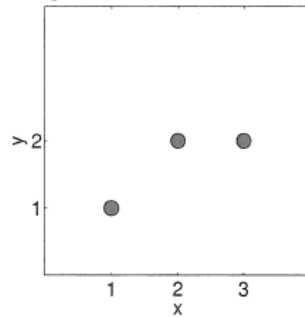
Section A

Quiz 1

Question 1: 4 marks

1) Compute the leave-one-out cross-validation error of a first-order (linear predictor).



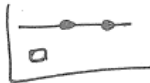


A: predictor 2 $\Rightarrow 1^2$

B: predictor $1\frac{1}{2}$ $\Rightarrow (\frac{1}{2})^2$

C: predictor 3 $\Rightarrow 1^2$

$$\Rightarrow \text{MSE} = \frac{1}{3} [1 + \frac{1}{4} + 1] = \frac{1}{3} \cdot \frac{9}{4} = \frac{3}{4}.$$



Question 2 : 3 marks

Suppose that we have training data $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$ and we wish to predict y using a nonlinear regression model with two parameters:

$$\hat{y} = a \log(x_1 + b)$$

We decide to train our model using gradient descent on the mean squared error (MSE).

Write down the expression for the MSE.

Just replace the expression of $\hat{y}$ according to the given $\hat{y}$

$$J(\theta) = \frac{1}{n} \sum_i (y^i - \hat{y}^i)^2 = \frac{1}{n} \sum_i (y^i - (a + \log(b x^i)))^2$$

Question 3: 3 marks

Which of the following are maximal item sets? Also, write down your reasoning for why you chose these item sets as maximal items using the Support threshold (by count): 4

		Items									
Transactions		A	B	C	D	E	F	G	H	I	J
	1										
	2										
	3										
	4										
	5										
	6										
	7										
	8										
	9										
	10										

Support threshold (by count) : 5

Maximal itemsets: {A}, {B}, {C}

Support threshold (by count): 4

Maximal itemsets: {A,B}, {A,C},{B,C}

Support threshold (by count): 3

Maximal itemsets: {A,B,C}

